INCH-POUND

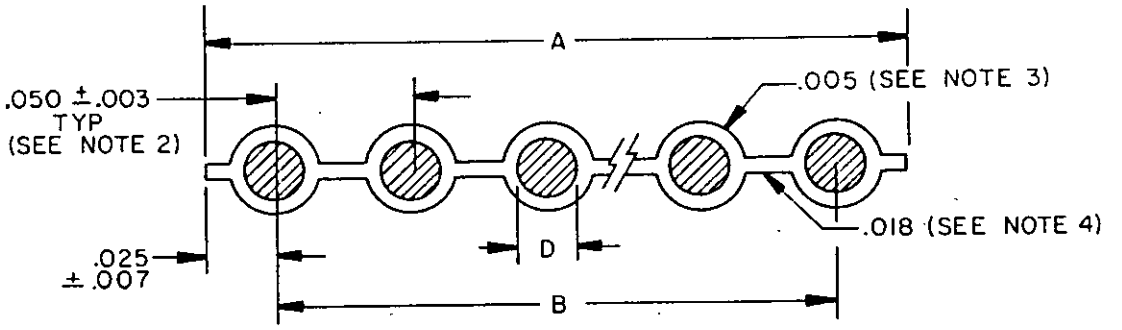
MIL-C-49055/12B
16 February 1990
SUPERSEDING
MIL-C-49055/12A
7 October 1983

MILITARY SPECIFICATION SHEET

CABLES, SPECIAL PURPOSE, ELECTRICAL, FLEXIBLE, FLAT, UNSHIELDED, ROUND CONDUCTOR, 200°C

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49055.



Inches	mm
.003	0.08
.005	0.13
.007	0.18
.018	0.46
.025	0.64
.050	1.27

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Minimum insulation thickness.
4. Maximum web thickness.

FIGURE 1. Round conductor flat cable.

(B) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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TABLE I. Dash numbers and characteristics.

Dash no.	No. of conductors	Cable width "A" inches	Cable thickness	Weight (lbs./ 1,000 ft.) Nominal 1/	Weight (lbs./ 1,000 ft.) Maximum	Conductor AWG	Conductor Dia "D" inches	Conductor Stranded	DC resistance at 20°C Ω / 1000 ft	End conductor to end conductor "B" inches
01	9	.450 ± .010	.029 ± .003	10.67	12.10	28	.015	7/36	65.9	.400 ± 0.007
02	10	.500 ± .010	"	12.30	14.00	"	"	"	"	.450 ± 0.007
02	10	.500 ± .010	"	12.30	14.00	"	"	"	"	.450 ± 0.007
03	14	.700 ± .010	"	17.52	19.99	"	"	"	"	.650 ± 0.007
04	15	.750 ± .010	"	19.15	21.89	"	"	"	"	.700 ± 0.007
05	16	.800 ± .011	"	20.14	22.99	"	"	"	"	.750 ± 0.011
06	20	1.000 ± .011	"	25.36	28.97	"	"	"	"	.950 ± 0.011
07	24	1.200 ± .011	"	30.29	34.57	"	"	"	"	1.150 ± 0.011
08	25	1.250 ± .011	"	31.92	36.47	"	"	"	"	1.200 ± 0.011
09	26	1.300 ± .011	"	33.20	37.95	"	"	"	"	1.250 ± 0.011
10	34	1.700 ± .015	"	43.64	49.91	"	"	"	"	1.650 ± 0.015
11	37	1.850 ± .015	"	47.25	54.01	"	"	"	"	1.800 ± 0.015
12	40	2.000 ± .015	"	51.48	58.89	"	"	"	"	1.950 ± 0.015
13	50	2.500 ± .015	"	62.88	72.21	"	"	"	"	2.450 ± 0.015
14	60	3.000 ± .020	"	77.60	88.81	"	"	"	"	2.950 ± 0.015
15	64	3.200 ± .020	"	81.72	93.42	"	"	"	"	3.150 ± 0.015
16	9	.450 ± .010	.032 ± .003	13.77	15.20	26	.019	7/34	41.2	.400 ± 0.007
17	10	.500 ± .010	"	15.40	17.10	"	"	"	"	.450 ± 0.007
18	14	.700 ± .010	"	21.86	24.33	"	"	"	"	.650 ± 0.007
19	15	.750 ± .010	"	23.80	26.54	"	"	"	"	.700 ± 0.007
20	16	.800 ± .011	"	25.10	27.95	"	"	"	"	.750 ± 0.011
21	20	1.000 ± .011	"	31.56	35.17	"	"	"	"	.950 ± 0.011
22	24	1.200 ± .011	"	38.04	42.14	"	"	"	"	1.150 ± 0.011
23	25	1.250 ± .011	"	39.67	44.24	"	"	"	"	1.200 ± 0.011
24	26	1.300 ± .011	"	41.26	46.01	"	"	"	"	1.250 ± 0.011
25	34	1.700 ± .015	"	54.18	60.45	"	"	"	"	1.650 ± 0.015
26	37	1.850 ± .015	"	58.71	65.48	"	"	"	"	1.800 ± 0.015
27	40	2.000 ± .015	"	63.88	71.29	"	"	"	"	1.950 ± 0.015
28	50	2.500 ± .015	"	80.04	89.37	"	"	"	"	2.450 ± 0.015
29	60	3.000 ± .020	"	96.20	107.41	"	"	"	"	2.950 ± 0.015
30	64	3.200 ± .020	"	101.56	113.26	"	"	"	"	3.150 ± 0.015
31	9	.450 ± .010	.038 ± .003	19.21	20.64	24	.024	7/32	25.4	.400 ± 0.007
32	10	.500 ± .010	"	20.84	22.54	"	"	"	"	.450 ± 0.007
33	14	.700 ± .010	"	29.48	31.95	"	"	"	"	.650 ± 0.007
34	15	.750 ± .010	"	31.96	34.70	"	"	"	"	.700 ± 0.007
35	16	.800 ± .011	"	33.80	36.65	"	"	"	"	.750 ± 0.011
36	20	1.000 ± .011	"	42.44	46.05	"	"	"	"	.950 ± 0.011
37	24	1.200 ± .011	"	51.64	55.94	"	"	"	"	1.150 ± 0.011
38	25	1.250 ± .011	"	53.27	57.84	"	"	"	"	1.200 ± 0.011
39	26	1.300 ± .011	"	55.40	60.15	"	"	"	"	1.250 ± 0.011
40	34	1.700 ± .015	"	72.68	78.95	"	"	"	"	1.650 ± 0.015
41	37	1.850 ± .015	"	78.84	85.60	"	"	"	"	1.800 ± 0.015
42	40	2.000 ± .015	"	85.64	93.05	"	"	"	"	1.950 ± 0.015
43	50	2.500 ± .015	"	107.24	116.57	"	"	"	"	2.450 ± 0.015
44	60	3.000 ± .020	"	128.84	140.05	"	"	"	"	2.950 ± 0.015
45	64	3.200 ± .020	"	136.37	148.06	"	"	"	"	3.150 ± 0.015

1/ Nominal values are for information only.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

(B) Insulation material: Fluorinated ethylene propylene (FEP); Polytetrafluoroethylene (PTFE); Perfluoral koxy (PFA).

Voltage rating: 300 volts rms at sea level.

Conductor coating: Silver.

Conductor stranding: See table I.

Weight: See table I.

Insulation flaws: 2,500 volts rms.

Dielectric withstanding voltage: 2,000 volts rms.

Conductor resistance: See table I.

(B) Temperature rating: -60°C to +200°C.

Thermal shock: -55°C ±5°C to +200°C ±5°C.

Flammability: Nonburning.

Flexing endurance: -55°C ±5°C to +200°C ±5°C for 24 AWG, 100 cycles; 26 AWG, 300 cycles; and 28 AWG, 500 cycles.

Moisture resistance:

Insulation resistance 50 megohms: 1,000 feet (minimum).

Low pressure (elevated temperature): +200°C ±5°C (maximum conductor temperature).

(B) Shrinkage: 0.125 inch maximum either end.

Markings (optional): In accordance with MIL-C-49055.

Strippability: Not applicable.

Solderability: Not applicable.

Tear groove propagation: Not applicable.

(B) Color: Optional.

(B) Part or Identifying Number (PIN): M49055/12- (dash number from table I). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification.

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Ⓑ TABLE II. Supersession data.

New PIN 1/	Superseded PIN 1/
M49055/12-**	M49055/12-**H M49055/12-**I M49055/12-**J M49055/12-**K

1/ ** denotes dash number from table I.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC, OS, SH
Air Force - 11, 17, 99
DLA - IS, ES
NASA - MSFC

User activities:
Army - AT, ME
Navy - MC

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1163-03)

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